

Dated: _____

QUESTION NO. 2

```
DECLARE Type, Mem, Dis, Avail : CHARACTER
DECLARE SOC, Req SOC, Hours, Time, Charge Needed,
Rate, Charge Discount, Charge Cost, Base Parking,
Parking Discount, Parking Cost, Total Discount, Total
Payable : Real
DECLARE Priority, Time status, stay message : STRING
```

```
Print ("Enter Vehicle Type (E for Electric, H for Hybrid)")
```

```
INPUT Type
```

```
Print ("Enter Current Battery SOC (1-100):")
```

```
INPUT SOC
```

```
Print ("Enter Expected parking duration (Hours)")
```

```
INPUT Hours
```

```
Print ("Enter Current time (24 hours)")
```

```
INPUT Time
```

```
Print ("Do you have a Parking membership? (Y/N)")
```

```
INPUT mem
```

```
Print ("Do you have disabled member priority? (Y/N)")
```

```
INPUT Dis
```

```
Print ("Is charging station available? (Y/N)")
```

```
INPUT Avail
```

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IF Avail == "N"

THEN

IF Type == "H" THEN

Print ("Charging Unavailable, Parking only").

ELSE

Print ("No charging available").

ENDIF

Charge cost = 0.0

ELSE

IF NOT (Type == "E" OR (Type == "H" AND
SOC < 40))

THEN

Print ("Vehicle does not qualify for
EV charging")

Charge cost = 0.0

ELSE

IF ReqSOC <= SOC THEN

Print ("No charging Required")

Charge cost = 0.0

ELSE

Charge Needed = ReqSOC - SOC

IF SOC <= 15 AND ReqSOC >= 80 THEN

Priority = "Emergency Charging priority (P1)"

ELSE IF Dis == "Y" OR (Mem == "Y" AND SOC <= 30)

THEN

Priority = "Priority charging (P2)"

ELSE

Priority = "Normal Charging (P3)"

ENDIF

Dated: _____

IF Time $\geq$ 17.00 AND Time $\leq$ 22.00

THEN

Time Status = "Peak"

Rate = 50.0

IF Mem = "Y" AND Priority != "Emergency Priority
(PI)"

THEN

Charge Discount = 0.10

ELSE

Charge Discount = 0.0

ENDIF

ELSE

Time status = "OFF - Peak")

Rate = 35.0

IF mem = "Y" AND Priority != "Emergency
Priority (PI)"

THEN

Charge Discount = 0.20

ELSE

Charge Discount = 0.0

ENDIF

ENDIF

Charge Cost = Charge Needed * Rate * (1.0 - charge Discount)

ENDIF

ENDIF

ENDIF

Dated: _____

IF Hours ≤ 2
THEN

Base Parking = 200

ELSE IF Hours ≤ 5

THEN

Base Parking = 400

ELSE

Base Parking = 700

ENDIF

ENDIF

IF Dis = "Y"

THEN

Parking Cost = 0.0

Parking Discount = Base Parking

ELSE IF Mem = "Y"

THEN

Parking Discount = Base Parking * 0.20

Parking Cost = Base Parking - Parking discount

ELSE

Parking Discount = 0.0

Parking cost = Base Parking

ENDIF

ENDIF

IF Hours > 8 THEN

Stay Message = "Long - stay warning: Please relocate
after charging")

Print (stay message)

ELSE

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Stay message = "Standard parking Duration"

Print (stay message)

ENDIF

Total Payable = Charge Cost + Parking cost

Total Discount = (Charge Needed * Rate * Charge Discount)
+ Parking Discount

Print ("Vehicle Type:" Type)

Print ("Current Battery SOC:" SOC, "%")

Print ("Required Battery SOC:" Req SOC, "%")

Print ("Charging Units needed:" Charge Needed, "%")

Print ("Charging Priority:" Priority)

Print ("Time Status:" Time Status)

Print ("Charging cost:" Charge cost)

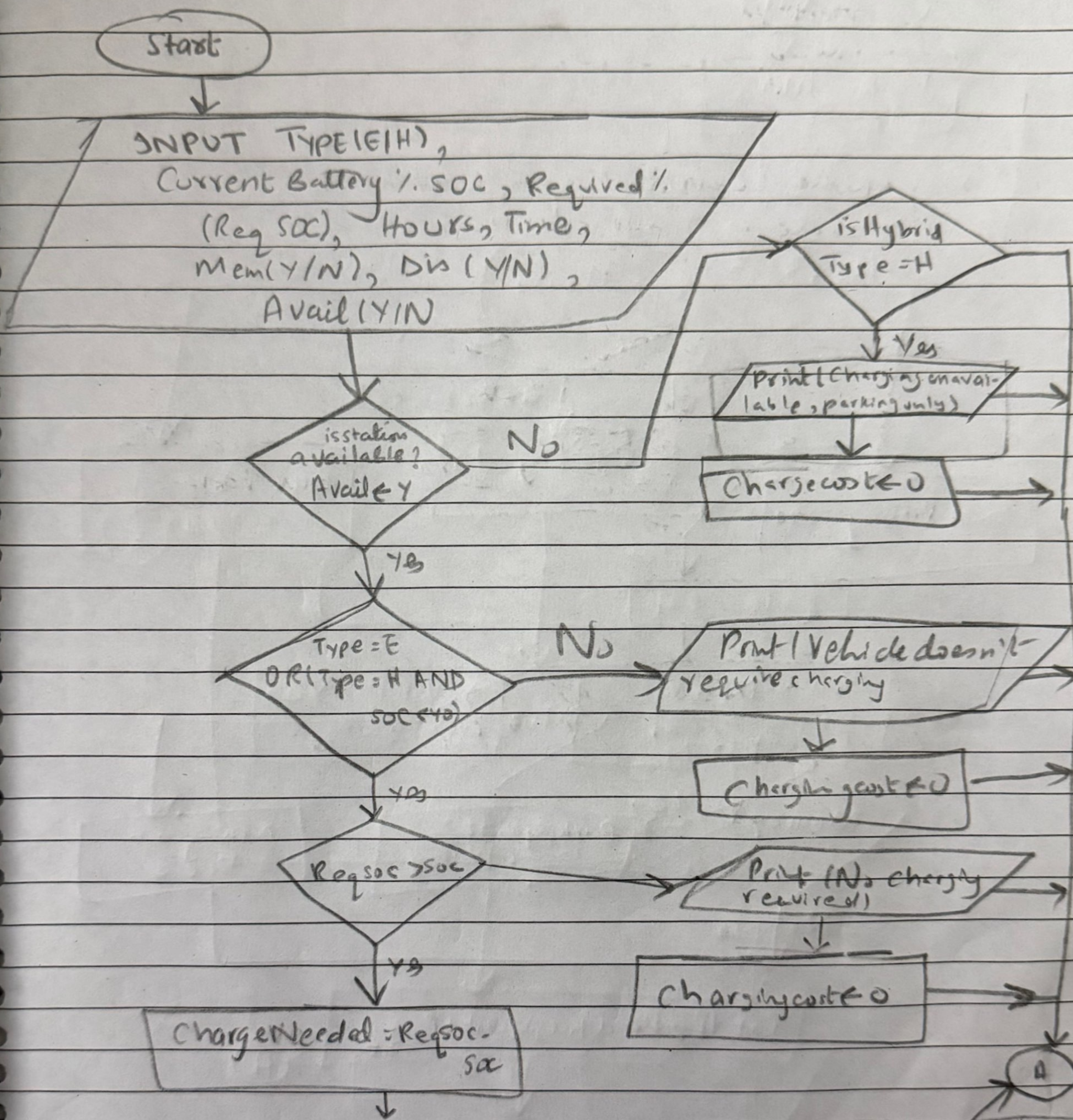
Print ("Parking cost:" Parking cost)

Print ("Discount:" Discount)

Print ("Payable amount:" Total Payable)

Print (stay message)

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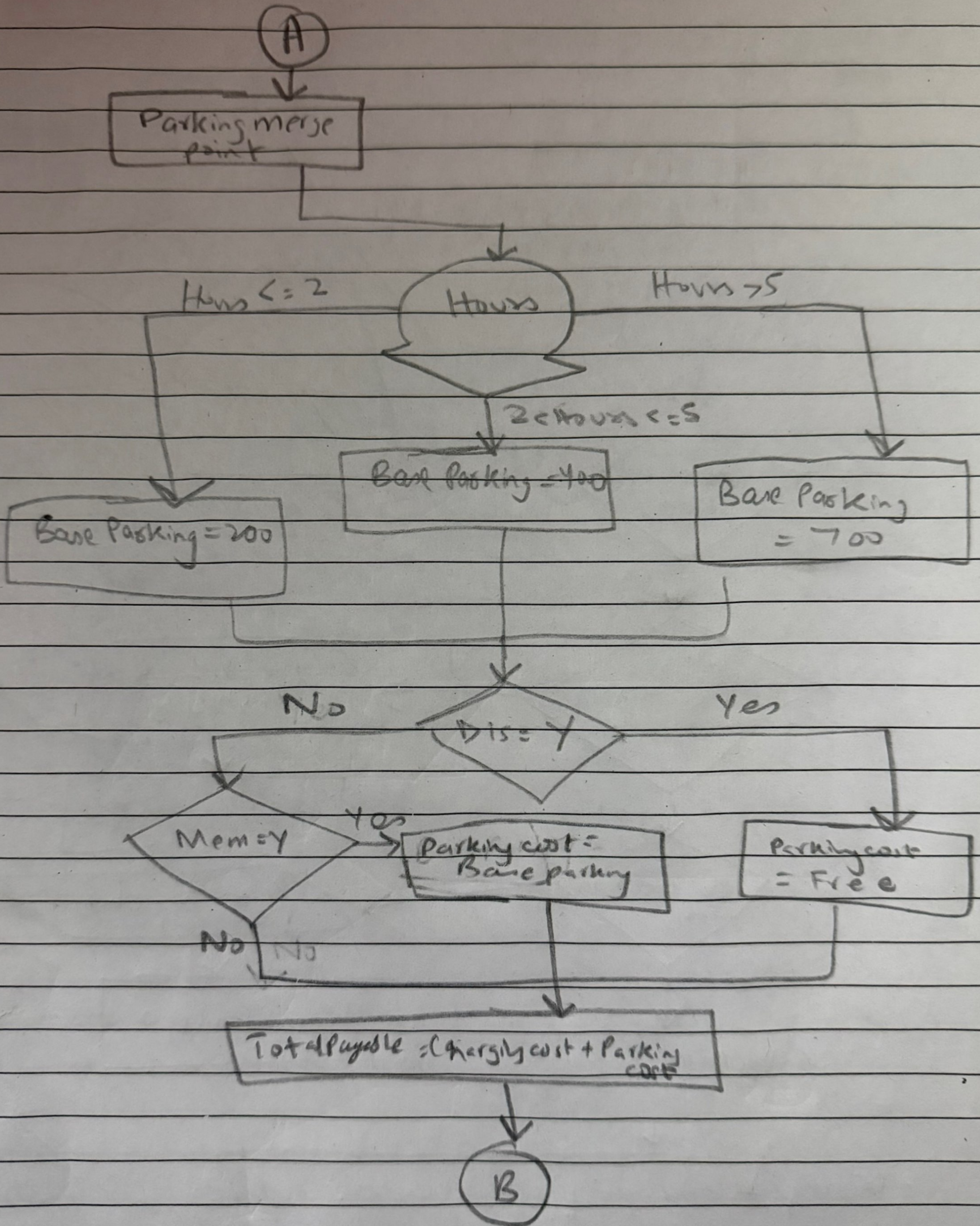


- Priority 1 (Emergency): If $SOC \leq 15\%$ and $Req\ SOC \geq 80\%$.

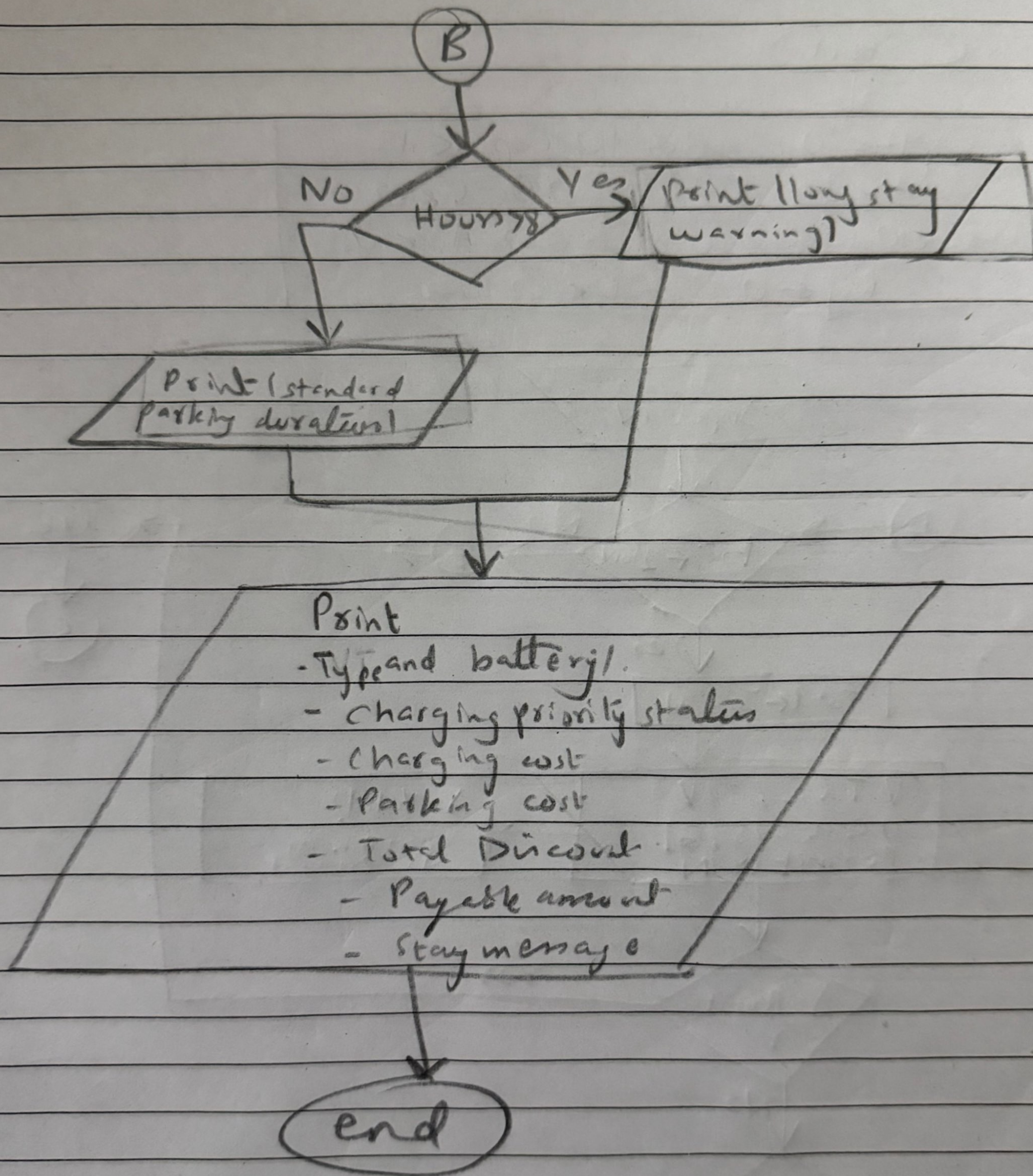
- Priority 2 (Priority): If disabled = Y or (Mem = Y and $SOC \leq 30\%$).

- Priority 3 (Normal): Otherwise:

Dated: _____



Dated: _____



INPUTS	OUTPUT
* Vehicle Type (Type: 'E' or 'H')	* Vehicle Type (Type)
* Current Battery SOC (SOC)	* Current Battery SOC (SOC)
* Target / Required SOC (ReqSOC)	* Required Battery SOC (ReqSOC)
* Expected Parking duration (Hours)	* Units needed (change needed)
* Current time (Time, 24-hrs format)	* Priority status message (Priority)
* Membership status (Mem: 'Y'/'N')	* Time category (Time Status)
* Disabled priority status (Dis: 'Y'/'N')	* Charging Cost (Charge Cost)
* Station availability (Avail: 'Y'/'N')	* Parking Cost (Parking Cost)
* Rates: Peak = 50.0, Off-Peak = 35.0	* Total Discount (Total Discount)
* Base Parking rates: 200 (≤ 2 hrs), 400 (≤ 5 hrs), 700 (> 5 hrs)	* Total payable (Total Payable)
	* Parking Stay message (StayMessage)
	* Status/error message (unavailability/qualification).

PROCESSING REQUIRED	SOLUTION ALTERNATIVES
* Check station availability (Av)	1) Calculate step-by-step using nested condition blocks (current pseudocode approach)
* Validate qualification: Vehicle must	2) Modularize into separate sub-

be Electric ('E) or Hybrid functions:

('H) with 50

* Calculate units needed:

Ch

* Determine Priority:

P1 (Emergency), P2 (Priority),

P3 (Normal)

* Determine Time Status

and Rate: Peak (17:00 -

22:00, 50.0) or Off-

Peak (35.0)

* Calculate Charging

Discount: 10% (Peak) or

20% (Off-Peak) if member

(non-P1)

* Calculate Ch

* Determine base parking

fee and apply disability /

membership discount

* Check stay warning

if Ho

* To

* To

Compute Charging Fee (),

Compute Parking Fee (), and

Display Bill ()

INPUT	PROCESSING	OUTPUT
* Type	1) Input Data: Read	* Type
* SOC	Type, SOC, Req SOC, Hours, Time, Mem, Dis, Avail	* SOC
* Req SOC	2) Check Availability & Eligibility:	* Req SOC
* Hours	* If Av, set Chc and display	* Charge Needed
* Time	unavailability message	* Priority
* Mem	* Else if not qualified (Ty, and	* Time Status
* Dis	not (Ty, and SOC), set Chc,	* Charge Cost
* Avail	* Else if Rec., no charge required	* Parking Cost
	(Chc).	* Total Discount
	* Else, Cha,	* Total Payable
	3) Determine Priority:	* Stay Message
	* If SO AND Rec	* status notific
	→ Priority: P1 (Emergency)	ation
	* Else if Di OR (Me AND SO) →	
	Priority: P2	
	(Priority).	
	* Else → Priority: P3 (Normal)	
	4) Determine Time status, Rate &	
	charging Discount:	
	* IF T ₁₁ AND T ₁₂ :	
	T ₁₁ , Ra)	
	If Me, and not P1, Ch, else	
	O.C,	
	* Else: T ₁₁ , Ra;	
	If Me, and not P1, Ch, else	

0. (,

* Ch,

5) Calculate Parking Cost & Parking Discount:

* Base Parking: 200 (Ho), 400 (Ho₁), 700 (Ho₂),

* If Di : Pa, Pa,

* Else if Me₁; Pa,
Pa;

* Else: Pa, Pa,

6) Check stay

Duration:

* If Ho, set warning stay messages;
else, standard duration message.

7) Calculate Totals:

* To

* To

8) Output Results:

Display all summary fields and bills